Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**4**
Electrons (Ne)	**4**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**4**
Eigenstate idx	**4 (1-indexed: 5)**

Eigenstate properties

Energy E0	**-0.82842712**
Energy E1	**0.00000000**
Gap (E1-E0)	**0.828427**
<S^2>	**2.000000**
S	**1.0000**
<Sz>	**0.000000**
Multiplicity	**3.0**
Spin label	**Triplet (S=1)**

Double occupancy

<D> total	**0.146447**
<D> / site	**0.036612**

RDM and GME statistics for this run (4-body clusters)

Hilbert space dim : 2^4 = 16
Number of clusters : 4
Pure clusters : 4
Mixed clusters : 0
Clusters with GME (>0) : 4

Purity gamma = Tr(rho^2) : mean = 1.000000 [min 1.000000 , max 1.000000]
Von Neumann entropy : mean = 0.000000 [min 0.000000 , max 0.000000]
Rank of rho : mean = 1.00 [min 1 , max 1]
C_GME (best estimate) : mean = 0.103857 [min 0.103857 , max 0.103857]

NOTE Analysis performed on eigenstate index 4 (state 5 in 1-indexed convention), not the ground state.

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Summary: All 4-Body RDM Clusters

System: L=4, Ne=4, U=4.0, t=1.0 | E_0=-0.828427

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_stiefel	C_UB	Status
(0, 1, 2, 3)	1.000000	0.103857	0.000000	0.000000	0.000000	0.103857	0.103857	GME
(1, 2, 3, 0)	1.000000	0.103857	0.000000	0.000000	0.000000	0.103857	0.103857	GME
(2, 3, 0, 1)	1.000000	0.103857	0.000000	0.000000	0.000000	0.103857	0.103857	GME
(3, 0, 1, 2)	1.000000	0.103857	0.000000	0.000000	0.000000	0.103857	0.103857	GME

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-1.149718e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(0,)	(1, 2, 3)	0.50000000
(0, 1)	(2, 3)	0.25269654
(0, 2)	(1, 3)	0.99460691
(0, 3)	(1, 2)	0.25269654
(0, 1, 2)	(3,)	0.50000000
(0, 1, 3)	(2,)	0.50000000
(0, 2, 3)	(1,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.10385652
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

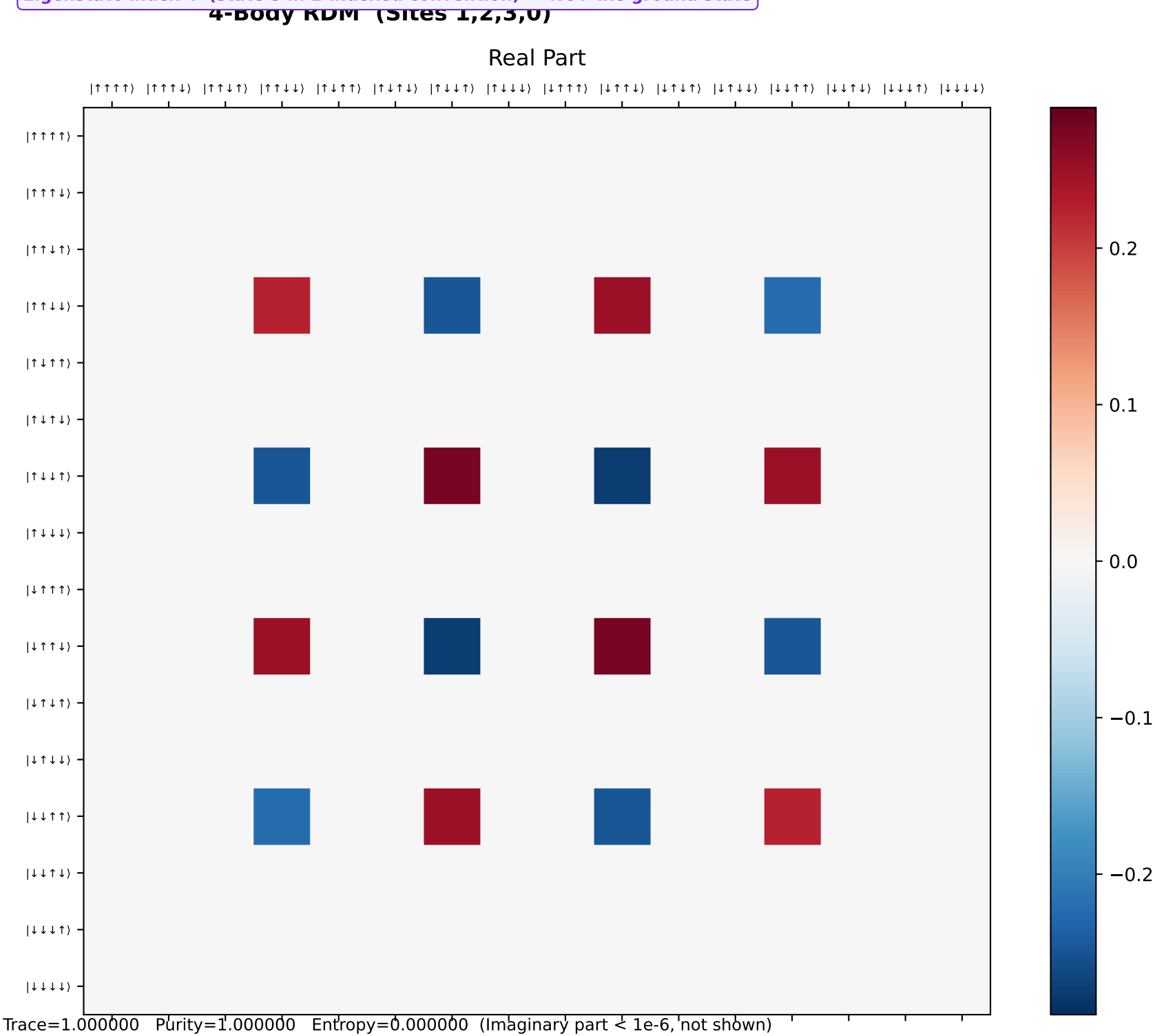
C_exact = 0.10385652
C_LB   = 0.55192826
Gap   = 0.44807174

Convex Roof Analysis
==========

Lower Bounds:
  C_LB   = 0.00000000
C_GME (unitary optimization):
  C_stiefel (unitary opt) = 0.10385652   BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.10385652

Status: nONE

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 1,2,3,0)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-6.332041e-17
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 0)	0.50000000
(1, 2)	(3, 0)	0.25269654
(1, 3)	(2, 0)	0.99460691
(1, 0)	(2, 3)	0.25269654
(1, 2, 3)	(0,)	0.50000000
(1, 2, 0)	(3,)	0.50000000
(1, 3, 0)	(2,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.10385652
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

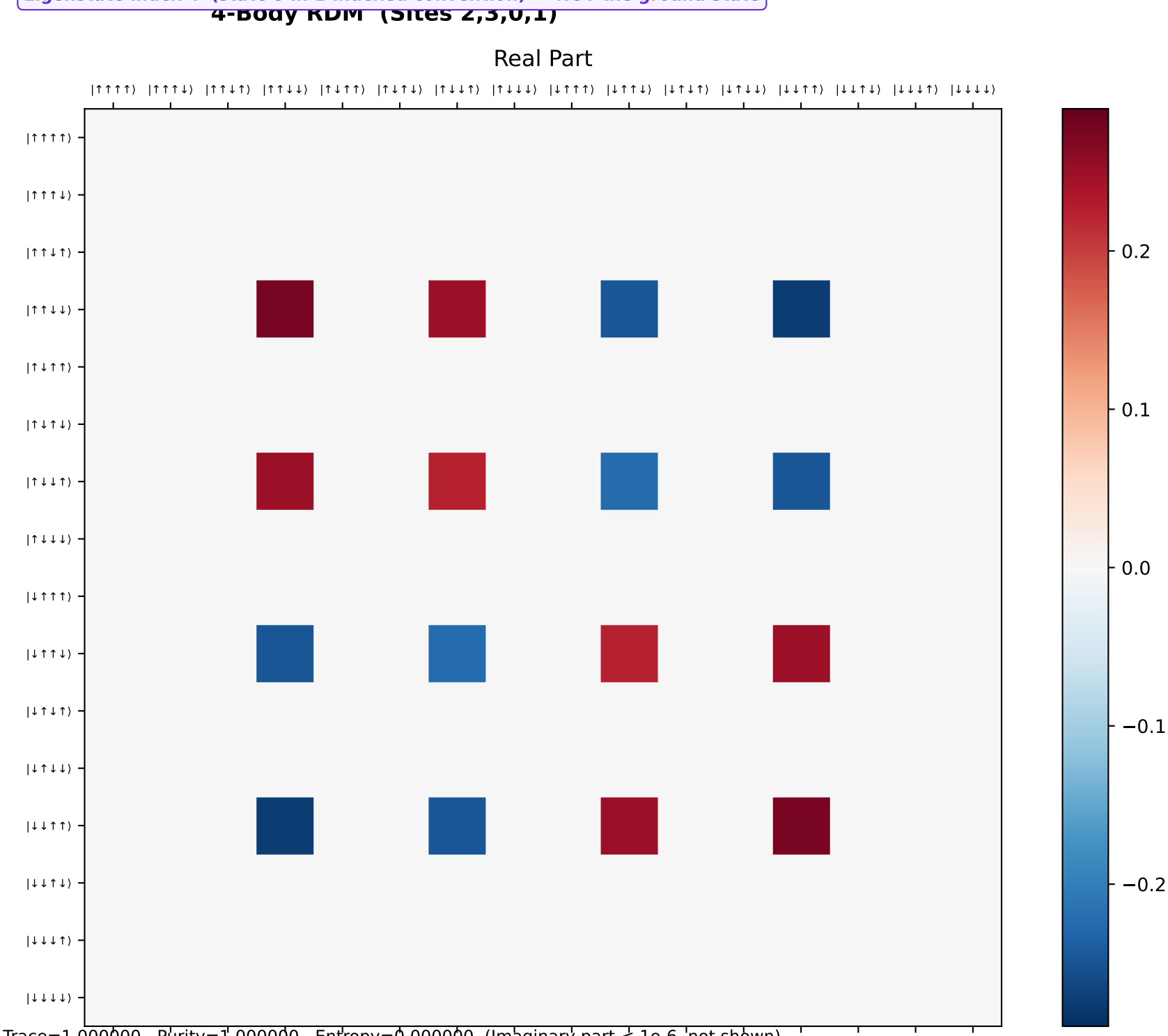
C_exact = 0.10385652
C_LB   = 0.55192826
Gap   = 0.44807174

Convex Roof Analysis
==========

Lower Bounds:
  C_LB   = 0.00000000
C_GME (unitary optimization):
  C_stiefel (unitary opt) = 0.10385652   BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.10385652

Status: nONE

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 2,3,0,1)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-1.467438e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(2,)	(3, 0, 1)	0.50000000
(2, 3)	(0, 1)	0.25269654
(2, 0)	(3, 1)	0.99460691
(2, 1)	(3, 0)	0.25269654
(2, 3, 0)	(1,)	0.50000000
(2, 3, 1)	(0,)	0.50000000
(2, 0, 1)	(3,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.10385652
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

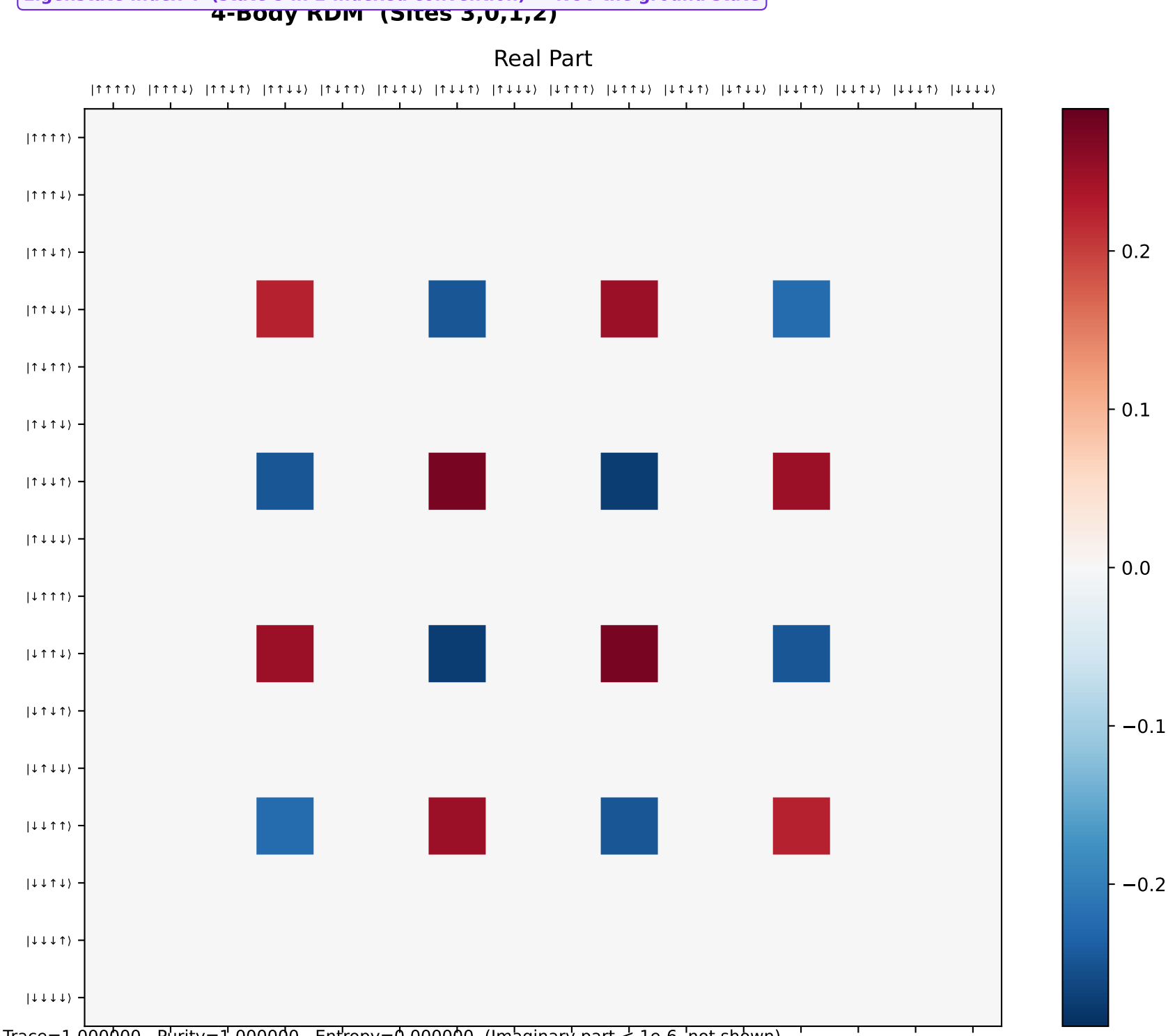
C_exact = 0.10385652
C_LB   = 0.55192826
Gap   = 0.44807174

Convex Roof Analysis
==========

Lower Bounds:
C_LB   = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = 0.10385652   BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.10385652

Status: nONE

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 3,0,1,2)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-1.828911e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(3,)	(0, 1, 2)	0.50000000
(3, 0)	(1, 2)	0.25269654
(3, 1)	(0, 2)	0.99460691
(3, 2)	(0, 1)	0.25269654
(3, 0, 1)	(2,)	0.50000000
(3, 0, 2)	(1,)	0.50000000
(3, 1, 2)	(0,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.10385652
GME confirmed across ALL cuts.
lower bound [eqs.6-9]

C_exact = 0.10385652
C_LB   = 0.55192826
Gap   = 0.44807174

Convex Roof Analysis
==========

Lower Bounds:
C_LB   = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = 0.10385652   BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.10385652

Status: nONE